REPORT

Project - Time-Series Statistical Analysis of Daily Temperature Trends.

Team – STATFORGE

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1. Introduction

Background & Motivation

In recent years, environmental conditions have become a major concern, especially in urban areas like Delhi where pollution levels fluctuate significantly. Two key indicators that help us understand these conditions are **temperature** and the **Air Quality Index (AQI)**.

Temperature affects not only weather patterns but also influences air pollution levels. Similarly, AQI provides a measure of how clean or polluted the air is, which directly impacts human health. However, these variables do not remain constant — they change continuously over time.

Because of this, simply looking at raw data is not enough. We need structured approaches like **statistical analysis and time-series techniques** to identify patterns, trends, and meaningful relationships.

This project is motivated by the need to:

- Understand how temperature and AQI behave over time
- Identify patterns that are not obvious at first glance
- Use data-driven methods to draw meaningful conclusions

Problem Context

Environmental data is inherently complex. Temperature and AQI values fluctuate daily due to various factors such as weather conditions, human activity, and seasonal changes.

When such data is viewed without analysis, it becomes difficult to:

- Recognize trends
- Understand relationships between variables
- Detect unusual or abnormal behavior

Traditional methods are not sufficient to extract insights from such continuously changing data. Therefore, there is a clear need for **systematic statistical and time-series analysis** to make sense of environmental patterns.

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2. Problem Statement

Temperature and air quality vary dynamically over time, making it challenging to analyze their behavior manually. Without proper analysis, important trends and relationships remain hidden.

The main problem addressed in this project is:

How can we use statistical and time-series techniques to analyze temperature and AQI data effectively and extract meaningful insights?

Core Gap

- Lack of structured analysis of environmental time-series data

- Difficulty in identifying trends and patterns manually
- Limited understanding of how temperature and AQI influence each other
- Absence of reliable methods for detecting anomalies in data

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3. Objectives

The main objectives of this project are:

- To analyze temperature and AQI data using statistical techniques
- To identify trends and patterns using time-series analysis
- To apply moving averages for smoothing fluctuations
- To study the relationship between temperature and AQI using correlation
- To detect anomalies using statistical methods
- To visualize data for better understanding and interpretation

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4. Literature Review

Several studies have explored the use of statistical and time-series methods for analyzing environmental data. These approaches help in identifying patterns, understanding variability, and improving data interpretation.

Study	Method Used	Key Findings
Study 1	Time-Series Analysis	Identified long-term temperature trends
Study 2	Statistical Measures	Used mean and standard deviation to analyze variation
Study 3	Correlation Analysis	Found relationships between environmental variables
Study 4	Moving Average	Reduced noise and highlighted patterns
Study 5	Visualization Techniques	Improved interpretation through graphs

These studies highlight the importance of combining statistical and visualization techniques for effective data analysis.

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5. Dataset Description

The dataset used in this project consists of **daily environmental data for Delhi over a period of 90 days**. It includes three main variables:

- **Temperature (°C)** – represents daily temperature
- **AQI** – indicates air quality levels
- **Humidity (%)** – shows moisture content in the air

The data is organized in a **time-series format**, where each entry corresponds to a specific date. This structure makes it suitable for analyzing trends over time.

Before analysis, the dataset was cleaned and processed to ensure:

- No missing values
- Consistent formatting
- Proper date conversion

This ensures that the analysis results are reliable and meaningful.

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6. Methodology

The project follows a structured approach:

1. Data Collection

The dataset was obtained and prepared for analysis.

2. Data Preprocessing

- Converted date into time-series format
- Cleaned and validated data

3. Descriptive Statistics

Basic statistical measures such as mean, median, and standard deviation were calculated to understand the distribution of data.

4. Time-Series Analysis

Line graphs were used to observe how temperature and AQI change over time.

5. Moving Average

Moving averages were applied to smooth short-term fluctuations and reveal underlying trends.

6. Correlation Analysis

The relationship between temperature and AQI was examined.

7. Outlier Detection

Z-score method was used to identify abnormal values in the dataset.

8. Visualization

Different types of graphs were used:

- Line plots
- Histograms
- Heatmaps
- Scatter plots

9. Trend Analysis

Regression techniques were used to identify overall trends in temperature data.

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7. SWOT Analysis

Strengths

- Uses real-world environmental data
- Combines statistical and time-series techniques
- Provides clear visual insights

Weaknesses

- Limited data duration
- Does not include all environmental factors

Opportunities

- Can be extended using machine learning models
- Can include multiple cities for comparison

Threats

- Data inaccuracies may affect results
- External factors may influence trends

Expected Outcomes

- Identification of trends and patterns
- Understanding relationship between variables
- Detection of anomalies

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8. Results and Discussion

Results

- Temperature shows a gradual increasing trend
- AQI shows a decreasing trend over time
- Moving averages smooth fluctuations and highlight trends
- Correlation analysis indicates a negative relationship between temperature and AQI

Discussion

The analysis reveals an interesting pattern: as temperature increases, AQI tends to decrease. This suggests an inverse relationship between the two variables.

Time-series analysis proved effective in identifying trends, while statistical methods helped in understanding variability. Visualization techniques made it easier to interpret complex data.

Evaluation Criteria (KPIs)

- Accuracy in identifying trends
- Strength of correlation between variables
- Effectiveness of smoothing techniques
- Ability to detect anomalies
- Clarity of data visualization

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9. Work Plan (PERT Chart)

Task	**Timeline**
Data Collection	February
Data Cleaning	March
Analysis	April
Report & Presentation	May

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10. Conclusion and Future Work

Conclusion

This project successfully demonstrates how statistical and time-series techniques can be used to analyze environmental data. The results highlight clear trends and relationships between temperature and AQI, providing meaningful insights.

Future Work

- Extend dataset to cover longer time periods
- Use machine learning for prediction
- Include additional environmental factors
- Perform comparative analysis across multiple cities

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11. References

- Kaggle Dataset
- Python Documentation (Pandas, NumPy, Matplotlib, Seaborn)
- Research papers on time-series analysis